Consider $\varphi \in C_0^2(\Omega) = \{\varphi \in C^2(\Omega),\ \varphi|_{\partial\Omega} = 0\}$, then if $u \in M_g \Rightarrow u + \varepsilon\varphi \in M_g$

let $f(\varepsilon) = F(u_0 + \varepsilon g)$, then $f(\varepsilon) \geq f(0) \quad \forall \varepsilon \in \mathbb{R}$

hence $f'(0) = 0 \Rightarrow f'(0) = \iint_\Omega Du_0 D\varphi \, dxdy = 0$

(Green's Formula: $\iint_\Omega \nabla u \cdot \nabla v \, dx = \iint_\Omega \operatorname{div}(v\nabla u) - v\Delta u \, dx$

$= \iint_{\partial\Omega} v \frac{\partial u}{\partial n} dS - \iint_\Omega v\Delta u \, dx$)

$\Rightarrow f'(0) \overset{\text{Green}}{=} \iint_{\partial\Omega} \varphi \cdot \frac{\partial u_0}{\partial n} dS - \iint_\Omega \varphi \Delta u_0 \, dx$

$= -\iint_\Omega \varphi \Delta u_0 dx \quad \Rightarrow -\Delta u_0 = 0$ (Laplace Eq.)

Rmk: There are two boundaries condition

Dirichlet B.C. : $u|_{\partial\Omega} = g$

Neumann B.C. : $\frac{\partial u}{\partial n}|_{\partial\Omega} = g$

} Robin B.C. : $\frac{\partial u}{\partial n} + a(x)u|_{\partial\Omega} = g$

We next study properties of harmonic function i.e. $u \in C^2(\Omega)$, $\Delta u = 0$

If u is a polynomial, then we call it a harmonic polynomial

Exp: $u=1$, $u=\sum_{k=1}^{n} c_k x_k + a$, $u = x_1^2 - x_2^2$, $u = \mathrm{Re}\, f(x+iy)$ if f is analytic on $\mathbb{C}$

① Mean value Formula

$$u(x) = \fint_{\partial B(x,r)} u(y)\, dS(y) = \fint_{B(x,r)} u(y)\, dy$$

Here: $\fint_{\partial B(x,r)} u(y)\, dS(y) = \frac{1}{n\alpha(n) r^{n-1}} \int_{\partial B(x,r)} u(y)\, dS(y)$, $\fint_{B(x,r)} u(y)\, dy = \frac{1}{\alpha(n) r^n} \int_{B(x,r)} u(y)\, dy$

Pf: Consider $\phi(r) = \fint_{\partial B(x,r)} u(y)\, dS(y) = \frac{1}{n\alpha(n) r^{n-1}} \int_{\partial B(x,r)} u(y)\, dS(y)$

$$= \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} u(x+rz)\, dS(z)$$

$$\phi'(r) = \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} \nabla u(x+rz) \cdot z\, dS(z)$$

$$\overset{\text{Div.}}{=} \frac{1}{n\alpha(n)} \int_{B(0,1)} \operatorname{div} u(x+rz)\, dz = \frac{1}{n\alpha(n)} \int_{B(0,1)} r \Delta u(x+rz)\, dz = 0$$

$\Rightarrow \phi(r) \equiv \text{const.} \Rightarrow \phi(r) = \phi(0) = u(x)$

Now : $\int_{B(x,r)} u(y)\,dy = \int_0^r \int_{\partial B(x,\theta)} u(y)\,dS(\theta)\,d\theta = \int_0^r \cdot\, u(x)\, n\alpha(n)\, \theta^{n-1} d\theta = u(x)\cdot\alpha(n)\cdot r^n$

$\Rightarrow \fint_{B(x,r)} u(y)\,dy = u(x)$ #

Conversely, we get Lemma 1:

Lemma 1: If $u\in C^2(\Omega)$ and satisfied the mean value formula, then u is harmonic.

pf: Use the notation of ϕ, we know $\phi'(r)=0$

Since $\forall x, r$, $\frac{1}{n\alpha(n)} \int_{B(0,1)} \Delta u(x+rz)\,dz = 0 \Rightarrow \Delta u = 0$

Lemma 2: If $u\in C(\Omega)$ and $\forall B(x,r)\subset\Omega$, $u(x) = \fint_{\partial B(x,r)} u(y)\,ds(y)$ then $u\in C^\infty(\Omega)$ and u is harmonic

Mollifier : $\eta(x) = \begin{cases} C e^{-\frac{1}{1-|x|^2}} & |x|<1 \\ 0 & |x|\geq 1 \end{cases}$, $\int_{\mathbb{R}^n} \eta(x)\,dx = 1$,

$\eta_\varepsilon(x) = \frac{1}{\varepsilon^n}\eta(\frac{x}{\varepsilon})$, $\operatorname{supp}\eta_\varepsilon(x) = \overline{B(0,\varepsilon)}$, $\int \eta_\varepsilon(x)\,dx = 1$

Proof of Lemma 2 : WLOG assume Ω is bounded

Let $u_\varepsilon(x) = \eta_\varepsilon * u\ (x) = \int_\Omega \eta_\varepsilon(x-y)\, u(y)\, dy$

Claim : $u_\varepsilon(x) \in C^\infty(\Omega)$ Check by DCT

now $\forall x \in \Omega_\varepsilon$, $u_\varepsilon(x) = \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)\, u(y)\, dy$

$$= \int_0^\varepsilon \int_{\partial B(x,r)} \eta_\varepsilon(x-y)\, u(y)\, dS(y)\, dr$$

$$= \int_0^\varepsilon \eta_\varepsilon(r) \int_{\partial B(x,r)} u(y)\, dS(y)\, dr$$

$$= u(x) \int_0^\varepsilon \eta_\varepsilon(r) \cdot |\partial B(x,r)|\, dr \quad = \int_{B(0,\varepsilon)} \eta_\varepsilon(x)\, dx = 1$$

$$= u(x) \quad \Rightarrow \quad u \in C^\infty(\Omega_\varepsilon) \quad \text{Let } \varepsilon \to 0^+ \quad \#$$

② Harnack Ineq.

Thm : $\Omega \subseteq \mathbb{R}^n$ open, V is connected compact subset of Ω, then there exists a constant $C = C(n, \Omega, V)$, s.t. $\forall$ nonnegative harmonic function u on Ω

$$\sup_V u \leq C \inf_V u$$

Rmk: Harnack Ineq. fails on Ω Ex: $\Omega = B(0,1)$, $u(x) = x_1 + 1$

Proof: Take $d = \operatorname{dist}(\partial\Omega, V)$, $r = \frac{1}{4}d$, $\forall x, y \in V$, $|x-y| < r$, Notice $B(x, 2r) \supset B(y, r)$

Then $u(x) = \frac{1}{\alpha(n)(2r)^n} \int_{B(x,2r)} u(z)\,dz \geq \frac{1}{\alpha(n)(2r)^n} \int_{B(y,r)} u(z)\,dz = \frac{1}{2^n} u(y)$

Cover V with open balls of radius r, $B_1 \cdots B_N$. $\Rightarrow \forall x, y$, $u(y) \leq u(x) \cdot 2^{nN}$

$\Rightarrow \sup_{y \in V} u(y) \leq 2^{nN} \inf_{x \in V} u(x)$

③ Strong Maximal Principle

Thm : Ω bounded open set in $\mathbb{R}^n$

$u \in C^2(\Omega) \cap C(\bar{\Omega})$ is harmonic on Ω, then :

a) $\max\limits_{\bar{\Omega}} u = \max\limits_{\partial\Omega} u$, $\min\limits_{\bar{\Omega}} u = \min\limits_{\partial\Omega} u$

b) If Ω is also connected, suppose we find $x_0 \in \Omega$, $u(x_0) = \max\limits_{\bar{\Omega}} u$

then u is constant.

Pf : For b) : take $B(x_0, r) \subset \Omega$, $u(x_0) = \frac{1}{\alpha(n) r^n} \int_{B(x_0,r)} u(y)\,dy \leq \frac{1}{\alpha(n) r^n} \int_{B(x_0,r)} u(x_0) = u(x_0)$

hence $\int_{B(x_0,r)} (u(x_0) - u(y))\,dy = 0 \Rightarrow u(y) \equiv u(x_0) \quad \forall\, y \in B(x_0, r)$

Consider $S = \{ z \in \Omega \mid u(z) = u(z_0) \}$ S open and close $\Rightarrow S = \Omega$

a) : Find the connected component containing x_0

Corollary 1: Ω is bounded open set, $f \in C(\Omega)$, $g \in C(\partial\Omega)$

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u|_{\partial\Omega} = g \end{cases} \quad (*)$$

$(*) \Rightarrow$ $(*)$ has at most one sol. $\in C^2(\Omega) \cap C(\bar{\Omega})$

Pf: Assume we have 2 solutions $u_1, u_2 \in C^2(\Omega) \cap C(\bar{\Omega})$ the $w = u_1 - u_2$ satisfies

$\Delta w = 0$ in Ω, $w|_{\partial\Omega} = 0 \Rightarrow w \equiv 0$ in Ω by Strong Maximal Principle.

hence $u_1 \equiv u_2$

Corollary 2: Ω connected & bounded open

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u|_{\partial\Omega} = g \end{cases} \qquad u \in C^2(\Omega) \cap C(\bar{\Omega})$$

If $g \geq 0$ then $u > 0$ in Ω

Furthermore, if we have $g(x_0) > 0$ for some $x_0 \in \partial\Omega$, then $u > 0$ in Ω.

④ Local estimate for harmonic functions

Thm: If u is harmonic in $\Omega \subseteq \mathbb{R}^n$, then $\forall B(x,r) \subset \Omega$, $\forall \alpha = (\alpha_1, \alpha_2, \cdots \alpha_n)$, $|\alpha| = \alpha_1 + \cdots + \alpha_n \leq k$

We have estimate $|D^\alpha u(x)| \leq \frac{C_k}{r^{n+k}} \int_{B(x,r)} |u(y)|\, dy$, $C_0 = \frac{1}{\alpha(n)}$, $C_k = \frac{(n+k)^{n+k}\ (n+1)^k}{\alpha(n)\ (n+1)^{n+1}}$

Pf: We prove it inductively

$k=0$: Mean Value Formular $\Rightarrow C_0 = \frac{1}{\alpha(n)}$

$k=1$: u harmonic $\Rightarrow$ u_{x_i} also harmonic

consider $0<s<r$: $u_{x_i}(x) = \frac{1}{\alpha(n)s^n} \int_{B(x,s)} u_{x_i}(y)\, dy$ Let $F = (0, \cdots \overset{i\text{th}}{\underset{\downarrow}{u}}, 0 \cdots 0)$

$u_{x_i} = \operatorname{div} F \Rightarrow \int_{B(x,s)} \operatorname{div} F\, dy = \int_{\partial B(x,s)} F \cdot \vec{n}\, ds(y) = \int_{\partial B(x,s)} u(y) \cdot n_i(y)\, ds(y)$

$\Rightarrow |u_{x_i}(x)| \leq \frac{1}{\alpha(n) \cdot s^n} \int_{\partial B(x,s)} |u(y)|\, dS(y)$

$\Rightarrow \int_0^r |u_{x_i}(x)|\, \alpha(n) \cdot s^n\, ds \leq \int_0^r \int_{\partial B(x,s)} |u(y)|\, dS(y) = \int_{B(x,r)} |u(y)|\, dy$

$\Rightarrow |u_{x_i}(x)| \cdot \frac{\alpha(n)}{n} r^{n+1} \leq \int_{B(x,r)} |u(y)|\, dy \Rightarrow |u_{x_i}(x)| \leq \frac{n}{\alpha(n)} \cdot \frac{1}{r^{n+1}} \int_{B(x,r)} |u(y)|\, dy$

$\Rightarrow C_1 = \frac{n}{\alpha(n)}$

Suppose we have estimate for $|\alpha| \le k-1$, consider $|\alpha| = k$, suppose $D^\alpha u = (D^\beta u)_{x_i}$ $|\beta| \le k-1$

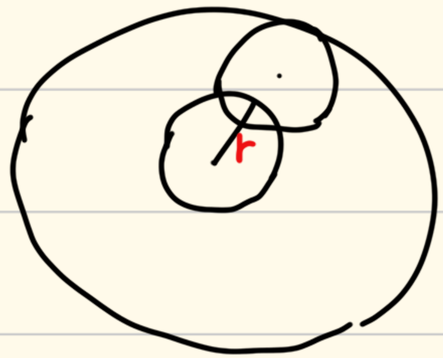
we consider ball $B(x, tr)$ $t \in (0,1)$

Repeat previous argument: $|D^\alpha u(x)| = |(D^\beta u)_{x_i}(x)| \le \frac{n}{\alpha(n)} \cdot \frac{1}{(tr)^{n+1}} \int_{B(x,tr)} |D^\beta u(y)|\, dy$

By induction: $|D^\beta u(y)| \le \frac{C_{k-1}}{((1-t)r)^{n+k-1}} \int_{B(y,(1-t)r)} |u(z)|\, dz$

$\Rightarrow |D^\alpha u(x)| \le \frac{1}{r^{n+k}} \frac{n}{t(1-t)^{n+k-1}} C_{k-1} \int_{B(x,r)} |u(z)|\, dz$

Let $t = \frac{1}{n+k}$, we get $C_k = C_{k-1} \cdot \frac{n}{\frac{1}{n+k} \cdot (1-\frac{1}{n+k})^{n+k-1}} = C_{k-1} \cdot \frac{(n+k)^{n+k}}{(n+k-1)^{n+k-1}} \cdot n$



#

⑤ Liouville's Thm

Thm: If u is harmonic on $\mathbb{R}^n$ & u bounded then u is constant.

Pf: $|Du| \le \frac{C_1}{r^{n+1}} \int_{B(x,r)} |u(y)|\,dy$, suppose $|u| \le k$

then $|Du| \le \frac{C_1 k}{r^{n+1}} \cdot \alpha(n) \cdot r^n = \frac{1}{r} \cdot \tilde{C}$, let $r \to +\infty$ #

⑥ Analyticity

Thm: u is harmonic in Ω then u is analytic

Pf: $\forall x \in \Omega$, $r = \frac{1}{4}\,\mathrm{dist}(x, \partial\Omega)$, we want to make u as a converging series in a nbhd of x_0

$$u(x) = \sum_{k=1}^{N-1} \sum_{|\alpha|=k} \frac{D^\alpha u(x_0)}{\alpha!} (x-x_0)^\alpha + R_N(x)$$

$$g(t) = u\left(x_0 + t(x-x_0)\right), \quad g(1) - g(0) = \sum_{k=1}^{N-1} \frac{g^{(k)}(0)}{k!} + \frac{g^{(N)}(t)}{N!}$$

$$g'(t) = (x-x_0)_i\, \partial_i u\left(x_0 + t(x-x_0)\right) = \left((x-x_0)\cdot\nabla\right) u\left(x_0 + t(x-x_0)\right)$$

$$g^{(k)}(t) = \left((x-x_0)\cdot\nabla\right)^k u\left(x_0 + t(x-x_0)\right) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} D^\alpha u\left(x_0 + t(x-x_0)\right) \cdot (x-x_0)^\alpha$$

hence we get $u(x) = \sum_{k=0}^{N-1} \sum_{|\alpha|=k} \frac{D^\alpha u(x_0)}{\alpha!} (x-x_0)^\alpha + \boxed{\sum_{|\alpha|=N} \frac{D^\alpha u(x_0 + t(x-x_0))}{\alpha!} (x-x_0)^\alpha}\ R_N(x)$

Recall facts : (i) $(1+\cdots+1)^N = \sum_{|\alpha|=N} \frac{N!}{\alpha!}$

(ii) Stirling's Formula : $\frac{N^{N+\frac{1}{2}}}{N!\, e^N} \to \frac{1}{\sqrt{2\pi}}$

Estimate $R_N(x)$: From Liouville : $z = x_0 + t(x-x_0) \in B(x_0, r) \Rightarrow B(z,r) \subset B(x, 2r)$

$$|D^\alpha u(z)| \le \frac{C_N}{r^{n+N}} \int_{B(z,r)} |u(y)|\, dy \le \frac{C_N}{r^{n+N}} \int_{B(x,2r)} |u(y)|\, dy$$

$$|R_n(x)| \le \sum_{|\alpha|=N} \frac{1}{\alpha!} \cdot \frac{C_N}{r^{n+N}} \cdot C \cdot |x-x_0|^N$$

$$= \frac{n^N}{N!} \cdot \frac{C_N}{r^{n+N}} \cdot C \cdot |x-x_0|^N \qquad \left(\text{if } \lambda\ C_N = \frac{1}{\alpha(n)} \cdot \frac{(n+N)^{n+N} \cdot (n+1)^N}{(n+1)^{n+1}}\right)$$

$$= \frac{C}{\alpha(n)(n+1)^{n+1} \cdot r^n} \cdot \frac{(n+N)^{n+N} (n+1)^N \cdot n^N}{r^N \cdot N!} |x-x_0|^N$$

$$= \tilde{C} \cdot \frac{(n+N)^{n+N}}{N!} \cdot \left(\frac{n(n+1)|x-x_0|}{r}\right)^N$$

$$\because (n+N)^{n+N} = N^N \cdot \left(1+\frac{n}{N}\right)^N \cdot (n+N)^n \sim N^N \cdot e^n \cdot (n+N)^n \sim \frac{1}{\sqrt{2\pi}} \cdot N! e^N \cdot \frac{e^n}{N^{\frac{1}{2}}} (n+N)^n$$

$$\Rightarrow |R_n(x)| \leq \tilde{C} \cdot \frac{N! e^N e^n (n+N)^n}{N^{\frac{1}{2}}} \cdot \frac{1}{N!} \cdot \left(\frac{(n+1)n|x-x_0|}{r}\right)^N$$

$$\leq \tilde{C}\ e^n \cdot (2N)^n \cdot \frac{e^N}{N^{\frac{1}{2}}} \cdot \left(\frac{(n+1)n|x-x_0|}{r}\right)^N$$

$$\leq \tilde{C}\ N^{n-\frac{1}{2}} \cdot \left(\frac{e(n+1)n|x-x_0|}{r}\right)^N$$

Take r_1, $\frac{e(n+1)nr_1}{r_0} \leq \frac{1}{2}$

then $\forall x \in B(x_0, r_1)$ $|R_n(x)| \leq C \cdot N^{n-\frac{1}{2}} \cdot (\frac{1}{2})^N$ let $N \to \infty$ ☒

Skipped Thm 2.10, 2.11 (Try it as Ex.)

2.2.3 Solving Laplace & Poisson Eq.

Case 1 : $\Omega = \mathbb{R}^n$, $-\Delta u = 0$

Let's seek for a solution $u = u(r)$, $r = |x|$

$$u_{x_i} = u'(r) \cdot \frac{x_i}{r}, \quad u_{x_i x_i} = u''(r) \cdot \frac{x_i^2}{r^2} + u'(r) \cdot \frac{1}{r} + u'(r) \cdot x_i \cdot \frac{-x_i}{r^3}$$

$$\Delta u = u''(r) + \frac{n-1}{r} u'(r) = 0 \quad \text{(Euler Eq.)}$$

Guess: $u(r) = r^\alpha \Rightarrow \alpha(\alpha-1) + (n-1)\alpha = 0 \Rightarrow \alpha = 0$ or $\alpha = -(n-2)$

$\alpha = 0 \rightsquigarrow u(r) = 1$, $\alpha = -(n-2) \rightsquigarrow u(r) = r^{2-n}$

If $n \geq 3$: we get $u(r) = a + b r^{2-n}$

$n = 2$: we get $u(r) = a + b \ln r$ } $r \neq 0$!

<u>This tell us</u> : there is no radically symetric func. that satisfy $-\Delta u = 0$ globally

Def: $\Gamma(x) = \begin{cases} a_2 \ln|x| & n=2 \\ a_n |x|^{2-n} & n\geq 3 \end{cases}$ we call it fundamental sol. to the Laplacian Eq. a_n 待定

Prop: ① $-\Delta\Gamma(x) = 0 \quad \forall x \neq 0$

② $|\alpha| = k > 1, \quad |D^\alpha \Gamma(x)| \leq C_k \cdot |x|^{2-n-k} \quad \forall x \neq 0$

Thm: $f \in C_c^2(\mathbb{R}^n)$

Consider $u(x) = \int_{\mathbb{R}^n} \Gamma(x-y) f(y)\, dy$, then $-\Delta u = f$ & $u \in C^2(\mathbb{R}^n)$

Pf: Step 1: Verify $u(x)$ is bounded

$$u(x) = \int_{\mathbb{R}^n} \Gamma(x-y) f(y)\, dy = \int_{B(x,\varepsilon)} \Gamma(x-y) f(y) + \int_{B(x,\varepsilon)^c} \Gamma(x-y) f(y)$$

$$\left|\int_{B(x,\varepsilon)} \Gamma(x-y) f(y)\, dy\right| \leq C \int_{B(x,\varepsilon)} |\Gamma(x-y)|\, dy = C \int_{B(0,\varepsilon)} |\Gamma(z)|\, dz = \begin{cases} n=2: & \tilde{C}\int_0^{\varepsilon}\int_0^{\varepsilon} \log r \cdot r\, dr = C(\varepsilon - \varepsilon^2 \log\varepsilon) \\ n\geq 3: & \tilde{C}\int_0^{\varepsilon} r^{2-n} \cdot r^{n-1}\, dr = C\varepsilon^2 \end{cases}$$

$$\left|\int_{B(x,\varepsilon)^c} \Gamma(x-y) f(y)\right| \leq \begin{cases} C\varepsilon^{2-n} & n\geq 3 \\ C(x) & n=2 \end{cases} \qquad \#$$

Step 2: Check u is C^2

$$u(x) = \int_{\mathbb{R}^n} I(y)\, f(x-y)\, dy \quad , \quad \frac{u(x+he_i)-u(x)}{h} = \int_{\mathbb{R}^n} I(y)\, \frac{f(x+he_i-y)-f(x-y)}{h}\, dy$$

$$= \int_{\mathbb{R}^n} I(y)\, \partial_i f(x-y+\theta he_i)\, dy \quad , \ 0 \le \theta \le 1$$

From $\partial_i f(x-y+\theta he_i) \xrightarrow{h\to 0} \partial_i f(x-y)$

and $|I(y)\, \partial_i f(x-y+\theta he_i)| \le |I(y)|\, C \cdot \chi_{\text{緊支}}$.

Hence, from DCT: $\lim_{h\to 0} \frac{u(x+he_i)-u(x)}{h} = \lim_{h\to 0} \int_{\mathbb{R}^n} I(y)\, \partial_i f(x-y+\theta he_i)\, dy \overset{\text{DCT}}{=\!=} \int_{\mathbb{R}^n} I(y)\, \partial_i f(x-y)\, dy$

Similarly, $\partial_i \partial_j u = \int_{\mathbb{R}^n} I(y)\, \partial_i \partial_j f(x-y)\, dy$

Step 3: Verify $-\Delta u = f$

(Green's 2nd Identity: $\int_{\mathbb{R}^n} f\Delta g - g\Delta f \, dx = \int_{\partial\Omega} f\frac{\partial g}{\partial n} - g\frac{\partial f}{\partial n} \, dS$

pf: $\int_\Omega f\Delta g \, dx = \int_\Omega \operatorname{div}(f\nabla g) - \nabla f\cdot\nabla g \, dx = \int_{\partial\Omega} f\,\nabla g\cdot\vec{n}\, dS - \int_\Omega \nabla f\nabla g\, dx$)

Hence:

$$-\Delta u = \int_{\mathbb{R}^n} \Gamma(y)(-\Delta_y) f(x-y)\,dy = \underbrace{\int_{B(x,\varepsilon)} \Gamma(y) - \Delta_x f(x-y)\,dy}_{(\mathrm{I})} + \underbrace{\int_{B(x,\varepsilon)^c} \Gamma(y) - \Delta_x f(x-y)\,dy}_{(\mathrm{II})}$$

$$(\mathrm{I}) \le C \int_{B(x,\varepsilon)} \Gamma(y)\,dy \to 0 \quad (\varepsilon\to 0)$$

$$(\mathrm{II}) = \underbrace{\int_{\partial B^c(x,\varepsilon)} f\frac{\partial \Gamma}{\partial n}}_{\mathrm{II}_1} - \underbrace{\Gamma\frac{\partial f}{\partial n}\,dS}_{\mathrm{II}_2}$$

$$|\mathrm{II}_2| \le C\int_{\partial B(0,\varepsilon)} |\Gamma(y)|\,dS(y) = \begin{cases} C\varepsilon|\log\varepsilon| & n=2 \\ C\varepsilon & n\ge 3 \end{cases} \to 0$$

$|\mathrm{II}_1|$: $n=2$ 时,

$$\mathrm{II}_1 = \int_{\partial B^c(x,\varepsilon)} f(x-y)\left(-\frac{a_r}{|y|}\right) dS(y) = \underbrace{\int_{\partial B^c(x,\varepsilon)} f(x)\left(-\frac{a_r}{|y|}\right) dS(y)}_{= f(x)(-2a_2\alpha(2)) \quad \to 0} + \int_{\partial B^c(x,\varepsilon)} \left(f(x-y)-f(x)\right)\left(-\frac{a_r}{|y|}\right) dS(y)$$

hence: $-\Delta u \to -2a_2\alpha(2)\cdot f(x)$ 进 $a_2 = -\frac{1}{2\pi}$

$n\geq 3$: $\mathrm{II}_1 = \underbrace{\int_{\partial B(x,\varepsilon)} \big(f(x-y)-f(x)\big)\cdot a_n(n-2)\,\varepsilon^{1-n}\,dS(y)}_{\to 0} + \underbrace{\int_{\partial B(x,\varepsilon)} f(x)\,a_n(n-2)\varepsilon^{1-n}\,ds(y)}_{\to f(x)\,a_n(n-2)\,n\alpha(n)}$

$\Rightarrow -\Delta u = f(x)\cdot\alpha(n)\,a_n(n-2)\,n \quad\Rightarrow\quad a_n = \dfrac{1}{\alpha(n)n(n-2)}$ #

Summarize : $\Gamma(x) = \begin{cases} -\frac{1}{2\pi}\cdot\ln|x| & n=2 \\ \dfrac{1}{\alpha(n)\,n(n-2)}\,|x|^{2-n} & n\geq 3 \end{cases}$

$-\Delta(\Gamma * f) = f$

Rmk: Finally introduce the "function" $\delta_0(x)$, $\int \delta_0(x) f(x)\,dx = f(x)$

Heuristic : $-\Delta\Gamma(x) = \delta_0(x)$, $-\Delta_x\Gamma(x-y) = 0 \quad (\forall y\neq x)$

考虑接地的导体边界，放入一点电荷，产生电势 $\Gamma(x-y)$ ← fundamental sol.

导体表面感应电荷重新分布，产生 $-h(x,y)$，而 $G(x,y)=\Gamma(x-y)-h(x,y)$ $\Rightarrow \begin{cases} -\Delta_x G(x,y)=\delta_y \\ G(x,y)=0 \end{cases}$

Step 1：解决点源问题：y为点源

$$\begin{cases} \Delta_x G(x,y)=\delta_y(x) & x \text{ in } \Omega \\ G(x,y)=0 & x \text{ in } \partial\Omega \end{cases}$$

$\Gamma(x-y)-h(x,y)$

Step 2：解决边界零值问题：

$$\begin{cases} -\Delta_x u=f(x) & \text{in } \Omega \\ u=0 & \text{in } \partial\Omega \end{cases}$$

then $u(x)=\int_\Omega f(y)\,G(x,y)\,dy$

Step 3：解决边界贡献：

$$\begin{cases} -\Delta_x u=f(x) & \text{in } \Omega \\ u=g & \text{in } \partial\Omega \end{cases}$$

then $u(x)=\int_\Omega f(y)\,G(x,y)\,dy+\int_{\partial\Omega} -\partial_{n_y} G(x,y)\,g(y)\,dS_y$

Case 2) : Ω is bounded region in $\mathbb{R}^n$

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{in } \partial\Omega \end{cases}$$

We can start with

$$\begin{cases} -\Delta u = f & x \in \Omega \\ u = 0 & x \in \partial\Omega \end{cases}$$

(Hope : Similar to $\Omega = \mathbb{R}^n$, we want to find $G(x,y)$. $\int_\Omega G(x,y) f(y)\,dy$ solve the eq.)

Such $G(x,y)$ is called Green - Function

Informally, take $f(x) = \delta_0(x-y)$, then

$$\begin{cases} -\Delta_x G(x,y) = \delta_0(x-y) & x \in \Omega \\ G(x,y) = 0 & x \in \partial\Omega \end{cases}$$

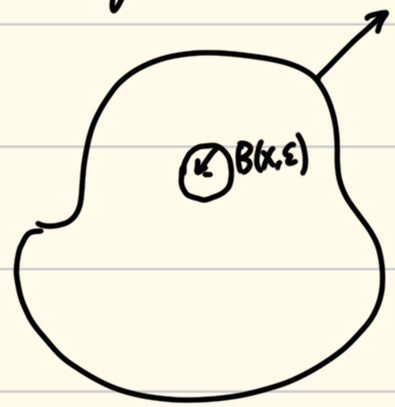
$\Gamma(x-y)$ solves 1 but fails at 2, Guess $G(x,y) = \Gamma(x-y) + \phi(x,y)$ → as a corrector

corrector ϕ satisfy :

$$\begin{cases} \Delta_x \phi(x,y) = 0 & x \in \Omega \\ \phi(x,y) = \Gamma(x-y) & x \in \partial\Omega \end{cases}$$

Suppose we find such $\phi(x,y)$. the we get a formula for Poisson eq.

Now we try to give rigorous derivation of Green's function and Poisson's Eq.



$$\Omega_\varepsilon = \Omega \setminus B(x,\varepsilon)$$

$$\int_{\Omega_\varepsilon} u(y)\,\Delta_y \Gamma(y-x) - \Delta_y u(y)\,\Gamma(y-x)\,dy$$

$$= \int_{\Omega_\varepsilon} u(y)\,\frac{\partial \Gamma}{\partial n}(y-x) - \frac{\partial u}{\partial n}(y)\cdot \Gamma(y-x)\,dS(y)$$

$$= \left(\int_{\partial\Omega} + \int_{\partial B(x,\varepsilon)}\right) u(y)\,\frac{\partial \Gamma}{\partial n}(y-x) - \frac{\partial u}{\partial n}(y)\cdot \Gamma(y-x)\,dS(y)$$

$$\left|\int_{\partial B(x,\varepsilon)} \frac{\partial u}{\partial n}\,\Gamma\,dS(y)\right| \begin{cases} \le C\int_{\partial B(x,\varepsilon)} \log\varepsilon\,dS(y) \le \widetilde{C}\,\varepsilon\log\varepsilon \\ \le C\int_{\partial B(x,\varepsilon)} \varepsilon^{2-n}\,dS(y) \le \overline{C}\,\varepsilon \end{cases} \to 0$$

$$n = -\frac{y-x}{|y-x|}, \quad \frac{\partial \Gamma}{\partial \vec{n}} = \frac{\partial \Gamma}{\partial y}\cdot\vec{n} = \begin{cases} -\frac{1}{2\pi}\cdot\frac{1}{|y-x|} & n=2 \\ \frac{1}{n\alpha(n)}\cdot\frac{1}{|y-x|^{n-1}} & n\ge 3 \end{cases} = \frac{1}{n\alpha(n)}\cdot\frac{1}{|y-x|^{n-1}}$$

$$\int_{\partial B(x,\varepsilon)} u\cdot\frac{1}{n\alpha(n)}\cdot\frac{1}{\varepsilon^{n-1}}\,dS(y) = \fint u\,dS(y) \to u(x)$$

How ?

Hence : let $\varepsilon\to 0$ $\quad u(x) = \int_\Omega f(y)\,\Gamma(y-x)\,dy - \int_{\partial\Omega} g(y)\,\frac{\partial \Gamma}{\partial n}\,dS(y) + \int_{\partial\Omega} \frac{\partial u}{\partial n}\,\Gamma\,dS(y)$ (Eq. 1)

To kill the last term we introduce the corrector $\phi^x(y)$:

$$\begin{cases} -\Delta_y \phi^x(y) = 0 & y \in \Omega \\ \phi^x(y) = \Gamma(x-y) & y \in \partial\Omega \end{cases}$$

Apply Green's 2nd Identity : $\int_\Omega \underbrace{u\Delta\phi}_{=0} - \phi\Delta u = \int_{\partial\Omega} u\frac{\partial\phi}{\partial n} - \underbrace{\phi}_{=\Gamma}\frac{\partial u}{\partial n}\, dS(y)$

$$\Rightarrow 0 = \int_{\partial\Omega} u\frac{\partial\phi}{\partial n} - \Gamma\frac{\partial u}{\partial n}\, dS(y) + \int_\Omega \phi\Delta u\, dy \quad (\text{Eq. } 2)$$

$$(\text{Eq. } 1) + (\text{Eq. } 2) \Rightarrow u(x) = \int_\Omega f(y)\Gamma(y-x)\,dy - \int_{\partial\Omega} g(y)\frac{\partial\Gamma}{\partial n}\,dS(y) + \int_{\partial\Omega} u\frac{\partial\phi}{\partial n}\,dS(y) - \int_\Omega \phi f\, dy$$

$$= \int_\Omega f(y)\left(\Gamma(y-x) - \phi^x(y)\right)dy - \int_{\partial\Omega} g\frac{\partial}{\partial n}\left(\Gamma(x-y) - \phi^x(y)\right)dS(y)$$

Summarize :

Def : $\begin{cases} -\Delta_y \phi^x(y) = 0 & y \in \Omega \\ \phi^x(y) = \Gamma(x-y) & y \in \partial\Omega \end{cases}$ we call $\phi^x(y)$ the corrector

Def : $G(x,y) = \Gamma(x-y) - \phi^x(y)$ the Green function on domain Ω

$G(x,y) : \bar{\Omega} \times \bar{\Omega} \setminus \{x=y\} \to \mathbb{R}$

$$\begin{cases} -\Delta_y G(x,y) = \delta(x-y) & y \in \Omega \\ G(x,y) = 0 & y \in \partial\Omega \end{cases}$$

Fact : $u(x) = \int_\Omega f(y)\, G(x,y)\, dy - \int_{\partial\Omega} g(y) \frac{\partial G}{\partial n} dy$ solves $\begin{cases} -\Delta u = f & x \in \Omega \\ u = g & x \in \partial\Omega \end{cases}$

Lemma: $\forall a,b\in\Omega,\ a\neq b,\ G(a,b)=G(b,a)$



Pf: Consider $f(x)=G(a,x),\quad g(x)=G(b,x)$

$$0=\int_{\Omega_\varepsilon}\Delta f\,g-f\,\Delta g\,dx$$

$$=\underbrace{\int_{\partial\Omega}\frac{\partial f}{\partial n}g-\frac{\partial g}{\partial n}f\,dS(x)}_{=0}+\int_{\partial B(a,\varepsilon)}\frac{\partial f}{\partial n}g-\frac{\partial g}{\partial n}f\,dS(x)-\int_{\partial B(b,\varepsilon)}\frac{\partial f}{\partial n}g-\frac{\partial g}{\partial n}f\,dS(x)$$

$$\int_{\partial B(a,\varepsilon)}\frac{\partial f}{\partial n}g\,dS(x)=\int_{\partial B(a,\varepsilon)}\underbrace{\frac{1}{n\alpha(n)\,\varepsilon^{n-1}}}_{\to g(a)}g\,dS(x)-\int_{\partial B(a,\varepsilon)}\underbrace{\frac{\partial\phi^x(y)}{\partial n}}_{\to 0}g(x)\,dS(x)$$

$$\left|\int_{\partial B(a,\varepsilon)}f\frac{\partial g}{\partial n}\,dS(x)\right|=\begin{cases}n=2: & \left|\int_{\partial B(a,\varepsilon)}\left(-\frac{1}{2\pi}\log\varepsilon-\phi^a(x)\right)\frac{\partial g}{\partial n}\,dS(x)\right|\leq C(\varepsilon\log\varepsilon+\varepsilon)\to 0\\ n\geq 3: & \left|\int_{\partial B(a,\varepsilon)}\left(\frac{1}{n\alpha(n)(2-n)}\,\frac{1}{\varepsilon^{n-2}}-\phi^a(x)\right)\frac{\partial g}{\partial n}\,dS(x)\right|\leq C(\varepsilon+\varepsilon^{n-1})\to 0\end{cases}$$

($\int_{\partial B(a,\varepsilon)}\frac{\partial f}{\partial n}g\,dS(x)\to 0$)

Hence, let $\varepsilon\to 0$: $g(a)=f(b)$ $\quad G(a,b)=G(b,a)$ #

Ex 1 : $\Omega = \mathbb{R}^n_+ = \{(x_1, \cdots x_n) \in \mathbb{R}^n,\ x_n > 0\}$ $\quad \partial\Omega = \{(x_1 \cdots x_{n-1}, 0)\} \cong \mathbb{R}^{n-1}$

We seek for $\phi^x(y)$ $\quad \begin{cases} -\Delta_y \phi^x(y) = 0 & y \in \mathbb{R}^n_+ \\ \phi^x(y) = \Gamma(x-y) & y \in \partial\mathbb{R}^n_+ \end{cases}$

$\forall x \in \mathbb{R}^n_+$, define $\hat{x} = (x_1 \cdots x_{n-1}, -x_n)$ be the reflection of x

$\Gamma(y-\hat{x}) = \Gamma(y_1 - x_1, y_2 - x_2, \cdots y_n + x_n)$ $\quad -\Delta_y \Gamma(y-\hat{x}) = 0$

$\Gamma(y-\hat{x})\big|_{y \in \partial\mathbb{R}^n_+} = \Gamma(x-y)$

hence we can take $\phi^x(y) = \Gamma(y-\hat{x})$, $G(x,y) = \Gamma(y-x) - \Gamma(y-\hat{x})$

We consider a special case $\quad \begin{cases} -\Delta u = 0 & x \in \mathbb{R}^n_+ \\ u = g & x \in \partial\mathbb{R}^n_+ \end{cases}$

then $u(x) = \int_{\partial\Omega} g(y) \frac{\partial G(x,y)}{\partial n} dS(y)$

$$\frac{\partial G}{\partial n} = -\frac{\partial G}{\partial y_n} = -\partial_{y_n} \Gamma(y-x) + \partial_{y_n} \Gamma(y-\tilde{x}) = \frac{(-1)}{n\alpha(n)} \frac{y_n - x_n}{|y-x|^n} - \frac{(-1)}{n\alpha(n)} \frac{y_n - \tilde{x}_n}{|y-\tilde{x}|^n}$$

$$\left.\frac{\partial G}{\partial y_n}\right|_{y \in \partial \mathbb{R}^n_+} = -\frac{1}{n\alpha(n)} \cdot \frac{x_n - \tilde{x}_n}{|y-x|^n} = \frac{2x_n}{n\alpha(n)\,|y-x|^n}$$

$$\Rightarrow u(x) = \int_{\partial \mathbb{R}^n_+} \frac{2x_n}{n\alpha(n)} \frac{g(y)}{|y-x|^n} \, dS(y) = \int_{\mathbb{R}^{n-1}} \frac{2x_n}{n\alpha(n)} \frac{g(y)}{\left[(y_1-x_1)^2 + \cdots + x_n^2\right]^{\frac{n}{2}}} \, dy$$

$$K(x,y) = \frac{2x_n}{n\alpha(n)\,|y-x|^n}, \quad u(x) = \int_{\mathbb{R}^{n-1}} K(x,y)\, g(y)\, dy$$

Thm: $g \in C_b(\mathbb{R}^{n-1})$ then $u = \int_{\partial \mathbb{R}^n_+} K(x,y)\, g(y)\, dS(y)$ satisfies

① $u \in C^\infty(\mathbb{R}^n_+)$ & bounded ② $-\Delta u = 0$

③ $\forall x_0 \in \partial \mathbb{R}^n_+$, $x \in \mathbb{R}^n_+$, $\lim_{x \to x_0} u(x) = g(x_0)$

Pf: ① By direct computation, $\int_{\partial \mathbb{R}^n_+} K(x,y)\, dS(y) = 1$, hence u is bounded

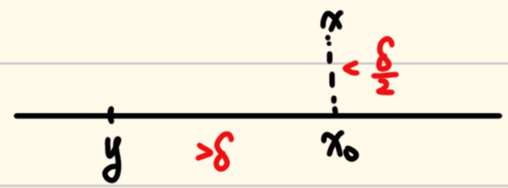
Also $K(x,y)$ is smooth w.r.t. x, hence $u \in C^\infty(\mathbb{R}^n_+)$

② $-\Delta_x K(x,y) = 0 \Rightarrow -\Delta u = 0$

③ Fix $x_0 \in \partial \mathbb{R}^n_+$, $\forall \varepsilon$, $\exists \delta_1 > 0$ s.t. $\forall |y - x_0| < \delta_1$, $y \in \partial \mathbb{R}^n_+$, $|g(y) - g(x)| < \frac{\varepsilon}{2}$

$$u(x) - g(x_0) = \int_{\partial \mathbb{R}^n_+} K(x,y)\,(g(y) - g(x_0)) = \int_{|y-x_0|<\delta} + \int_{|y-x_0|>\delta}$$

$$\left|\int_{|y-x_0|<\delta}\right| \le \frac{\varepsilon}{2} \int_{\partial \mathbb{R}^n_+} K(x,y) = \frac{\varepsilon}{2} \qquad \left|\int_{|y-x_0|>\delta}\right| \le C \int_{|y-x_0|>\delta} K(x,y)\, dy$$



若 $|x - x_0| < \frac{\delta_1}{2}$, 则 $|y - x| \ge \frac{1}{2}|y - x_0| \Rightarrow \left|\int_{|y-x_0|>\delta}\right| \le C \int_{|y-x_0| \ge \delta_1} \frac{x_n}{\left(\frac{1}{2}|y-x_0|\right)^n}\, dy \le \tilde{C} \int_{r \ge \delta_1} \frac{x_n}{r^n} \cdot r^{n-2}\, dr = \tilde{C}\, x_n \frac{1}{\delta_1} \le \frac{\varepsilon}{2}$

取 $x_n \le \frac{\varepsilon \delta_1}{2\tilde{C}}$

取 $|x - x_0| \le \delta = \min\left\{\frac{\delta_1}{2}, \frac{\varepsilon \delta_1}{2\tilde{C}}\right\}$, $\left|\int_{|y-x_0|>\delta}\right| \le \frac{\varepsilon}{2}$ $\Rightarrow u(x) \to g(x_0)$

Section 2.2 Laplace Eq.

Ex2 $\Omega = B(0,R)$ $\begin{cases} -\Delta u = f & x \in \Omega \\ u|_{\partial\Omega} = g \end{cases}$ First consider $R=1$

Search for corrector : $\begin{cases} -\Delta_y \phi^x(y) = 0 & \text{in } B(0,1) \\ \phi^x(y) = \Gamma(x-y) & \text{in } \partial B(0,1) \end{cases}$

Consider $x^* = \frac{x}{|x|^2}$, $x^* \cdot x = 1$

Take $\phi^x(y) = \Gamma\left(|x|(y-x^*)\right)$. $G(x,y) = \Gamma(y-x) - \Gamma\left(|x|(y-x^*)\right)$

So solution $u(x) = -\int_{\partial B_1} g(y) \frac{\partial G(x,y)}{\partial n_y} dS_y$

Compute : $\frac{\partial G}{\partial n_y} = \frac{\partial G}{\partial y} \cdot \frac{y}{|y|} = \frac{1}{n\alpha(n)} \left(\frac{x-y}{|y-x|^n} - \frac{x^*-y}{|x^*-y|^n} \cdot \frac{|x|^2}{|x|^n} \right) \cdot \frac{y}{|y|}$ $(n \geq 3)$

$\Rightarrow \left.\frac{\partial G}{\partial n_y}\right|_{|y|=1} = \frac{1}{n\alpha(n)} \frac{(|x|^2-1)}{|y-x|^n}$

$\Rightarrow u(x) = -\int_{\partial B_1} \frac{1}{n\alpha(n)} \frac{1-|x|^2}{|y-x|^n} g(y)\, dS(y)$

Back to $B(0,R)$: Thm: $g: \partial B(0,R) \to \mathbb{R}$ continuous function, $u = \int_{\partial B(0,R)} K_R(x,y)\, g(y)\, dS(y)$ satisfy

① u bounded & C^∞ ② $-\Delta u = 0$ in $B(0,R)$

Here: $K_R(x,y) = \dfrac{R^2 - |x|^2}{n\alpha(n) R |y'-x'|^2}$

2.2.4 Estimates

a) L^∞ - estimate

Thm: $\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{in } \partial\Omega \end{cases}$, u is a $C^2(\Omega) \cap C(\bar{\Omega})$, then we have

$\sup_{x \in \bar{\Omega}} |u(x)| \le \sup_{\partial\Omega} |g| + C \sup_{\Omega} |f|$ C is a constant of dim & diam(Ω)

Pf: Easy case. $f=0$: Apply maximal principle.

Let's find a function $v(x)$, $-\Delta v = 1$

Put Ω in a large ball $B_d(x_0)$, consider $v(x) = \frac{1}{2n}(d^2 - |x-x_0|^2) \le \frac{d^2}{2n}$

Denote $F = \sup|f|$, $G = \sup|g|$ and consider $w_1(x) = u(x) - G - Fv(x)$

Then $\begin{cases} -\Delta w_1(x) = -\Delta u + F\Delta v(x) = f - F \le 0 \\ w_1(x)|_{\partial\Omega} = g - G - Fv(x)|_{\partial\Omega} = g - G \le 0 \end{cases}$

Since subharmonic func. reaches max on bdry.

$\Rightarrow w_1(x) \le 0 \Rightarrow u(x) \le G + Fv(x) \le G + F \cdot \frac{d^2}{2n}$

Similarly, consider $w_2(x) = -u(x) - G - Fv(x) \Rightarrow w_2(x) \le 0 \Rightarrow u(x) \ge -G - F \cdot \frac{d^2}{2n}$

Hence $\sup|u(x)| \le G + F \cdot \frac{d^2}{2n}$

Corollary : $\begin{cases} -\Delta u_i = f_i \\ u_i = g_i \end{cases}$ then $\sup |u_i - u_j| \le \sup |g_i - g_j| + C \sup |f_i - f_j|$

b) Energy Estimate

Thm : $\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{in } \partial\Omega \end{cases}$ $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ is a sol.

Then : $\int_\Omega |Du|^2 dx \le C \int_\Omega |f|^2 dx$, $C = C(\dim, \operatorname{diam}(\Omega))$

Pf : $\int_\Omega -\Delta u \cdot u = \int_\Omega fu$, $\nabla \cdot (u \nabla u) = |\nabla u|^2 + u \Delta u$

$\Rightarrow \int_\Omega |Du|^2 dx - \int_\Omega \nabla \cdot (u \nabla u) dx = \int_\Omega fu$

$\Rightarrow \int_\Omega |Du|^2 dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} u \, dS = \int_\Omega fu$

$\Rightarrow \int_\Omega |Du|^2 dx = \int_\Omega fu \, dx \le \frac{\varepsilon}{2} \int_\Omega u^2 dx + \frac{1}{2\varepsilon} \int_\Omega f^2 dx$

Lemma (Poincare Ineq.) Ω bounded

① $u \in C_0^1(\Omega) = \{ u \in C^1(\Omega),\ \text{supp } u \text{ compact and } \subset \Omega \}$

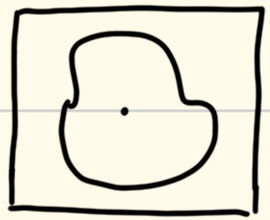
then $\int_\Omega |u|^2 dx \leq C \int_\Omega |Du|^2 dx$

② $u \in C^1(\bar{\Omega})$ & $u = 0$ in $\partial\Omega$

then $\int_\Omega |u|^2 dx \leq C \int_\Omega |Du|^2 dx$

Pf of Lemma :

① extend $\tilde{u} = \begin{cases} u & x \in \bar{\Omega} \\ 0 & x \notin \Omega \end{cases}$ $\tilde{u} \in C^1(\mathbb{R}^n)$



put Ω in a large box D, WLOG $D = [-d, d]^n$ by assumming $0 \in \Omega$

$\forall\ x = (x_1, \cdots x_n) \in \Omega$

$$\tilde{u}(x_1, \cdots x_n) = \tilde{u}(x_1, \cdots x_n) - \tilde{u}(x_1, \cdots x_{n-1}, -d) = \int_{-d}^{x_n} \partial_{x_n} \tilde{u}(x_1 \cdots x_{n-1}, \xi)\, d\xi$$

$$\Rightarrow |\tilde{u}(x_1, \cdots x_n)|^2 \overset{C\text{-}S}{\leq} \int_{-d}^{x_n} 1\, d\xi \cdot \int_{-d}^{x_1} (\partial_{x_n} \tilde{u})^2 (x_1 \cdots x_{n-1}, \xi)\, d\xi \leq 2d \int_{-d}^{d} (\partial_{x_n} \tilde{u})^2 (x_1 \cdots x_{n-1}, \xi)\, d\xi$$

Hence by Fubini $$\int_{-d}^{d}\cdots\int_{-d}^{d}|\tilde{u}(x)|^2dx=\int_{-d}^{d}\cdots\int_{-d}^{d}\left(\int_{-d}^{d}|\tilde{u}(x_1,\cdots x_{n-1},x_n)|^2dx_n\right)dx'$$

$$\leq 2d\int_{-d}^{d}\cdots\int_{-d}^{d}\left(\int_{-d}^{d}|\partial_{x_n}\tilde{u}(x_1,\cdots x_{n-1},\xi)|^2d\xi\right)dx_n\,dx'$$

$$=(2d)^2\int_{-d}^{d}\cdots\int_{-d}^{d}|\partial_{x_n}\tilde{u}(x_1,\cdots x_{n-1},\xi)|^2d\xi\,dx'$$

$$\leq(2d)^2\int_{\Omega}|Du|^2dx$$



② $\forall\,(x_1\cdots x_{n-1})=x'\in\mathbb{R}^{n-1}$

$I_{x'}=\{x_n\in\mathbb{R}\mid(x_1\cdots x_{n-1},x_n)\in\Omega\}\subseteq\mathbb{R}$ and $I_{x'}$ is open!

$I_{x'}=\bigcup_{j=1}^{\infty}(a_j,b_j)$

since $u(x',t)-u(x',a)=\int_{a_j}^{t}\partial_{x_n}(x',\xi)d\xi\overset{C.S.}{\Rightarrow}|u(x',t)|^2\leq d\int_{a_j}^{t}(\partial_{x_n}u)^2(x',\xi)\,d\xi$

$$\Rightarrow\int_{a_j}^{b_j}|u(x',t)|^2dt\leq d\int_{a_j}^{b_j}\int_{a_j}^{t}|\partial_{x_n}u(x',\xi)|^2d\xi\,dt\leq d^2\int_{a_j}^{b_j}|\partial_{x_n}u(x',\xi)|^2d\xi$$

$$\Rightarrow\int_{I_{x'}}|u(x',t)|^2dt\leq d^2\int_{I_{x'}}|\partial_{x_n}u(x',t)|^2dt$$

(Fubini) $$\Rightarrow\int_{\Omega}|u|^2dx\leq d^2\int_{\Omega}|Du|^2d\lambda$$

Back to thm: $\int_\Omega |Du|^2 dx = \int_\Omega fu\, dx \leq \frac{\varepsilon}{2}\int_\Omega u^2 dx + \frac{1}{2\varepsilon}\int_\Omega f^2 dx$

$\leq \frac{\varepsilon C}{2}\int_\Omega |Du|^2 dx + \frac{1}{2\varepsilon}\int_\Omega f^2 dx$

Pick $\varepsilon = \frac{1}{C} \Rightarrow \int_\Omega |Du|^2 dx \leq C\int_\Omega f^2 dx$

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